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Inventor(s)

Page; Joey

Retractable vehicle mud flap arrangement

Abstract

A retractable vehicle mud flap arrangement includes a drive unit, an axle, mud flaps, and a control unit. The axle is operatively connected to the drive unit and designed to be rotated by the drive unit. The mud flaps are suspended from opposing ends of the axle. The control unit is operatively connected to the drive unit and designed to control the operation of the drive unit. The drive unit is actuatable to rotate the drive axle and roll up and retract the mud flaps from an extended position to a retracted position to minimize contact with objects on the ground and damage resulting from such contact upon a vehicle is driven in reverse.

Inventors: Page; Joey (Burleson, TX)

Applicant: Page; Joey (Burleson, TX)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

[0003] Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM.

[0004] Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR

[0005] Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

[0006] The disclosure relates to vehicle mud flaps and more particularly pertains to a new retractable vehicle mud flap arrangement for preventing damage to the mud flaps. Some vehicles, such as trucks or tractor trailers, require mud flaps approved by the Department of Transportation (DOT) to promote highway safety. When driving, it is not uncommon for mud flaps to be damaged or torn off, resulting in costly and time-consuming repairs or replacement. By making the mud flaps part of a retractable system, the mud flaps could be retracted and thereby protected from damage. To further explain, while mud flaps often experience normal wear and tear from objects during driving, damage can occur when driving the vehicle in reverse. The mud flaps can contact objects on the ground, such as curbs or parking blocks, for example. They can snag and be torn, or possibly become trapped between the vehicle tires and the objects and be damaged or destroyed as the vehicle backs up. If the mud flaps are retractable, especially during driving in reverse, the mud flaps can be raised a distance sufficient to avoid such damaging objects.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98.

[0007] The prior art relates to vehicle mud flaps. The prior art, as best understood, does not disclose a retractable mud flap system for retracting the mud flaps, such as when driving in reverse.

BRIEF SUMMARY OF THE INVENTION

[0008] An embodiment of the disclosure meets the needs presented above in a retractable vehicle mud flap arrangement generally comprising a drive unit, an axle, mud flaps, and a control unit. The drive unit is designed to be mounted adjacent wheels of a vehicle. The axle is operatively connected to the drive unit and designed to be rotated by the drive unit. The mud flaps are suspended from opposing ends of the axle. The control unit is operatively connected to the drive unit and designed to control the operation of the drive unit. The drive unit is actuatable to rotate the drive axle and roll up and retract the mud flaps from an extended position to a retracted position to minimize contact with objects on the ground and damage resulting from such contact upon a vehicle is driven in reverse.

[0009] There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

[0010] The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

[0011] The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

[0012] FIG. **1** is a rear view of a retractable vehicle mud flap arrangement according to an embodiment of the disclosure.

[0013] FIG. **2** is a rear view of an embodiment of the disclosure.

[0014] FIG. **3** is a partial schematic view of an embodiment of the disclosure.

[0015] FIG. **4** is a side view of a portion of an embodiment of the disclosure.

[0016] FIG. **5** is a rear view of a portion of an embodiment of the disclosure.

[0017] FIG. **6** is a side view of a portion of an embodiment of the disclosure.

[0018] FIG. **7** is an exploded view of a portion of an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0019] With reference now to the drawings, and in particular to FIGS. **1** through **7** thereof, a new retractable vehicle mud flap arrangement embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral **10** will be described.

[0020] As best illustrated in FIGS. **1** through **7**, the retractable vehicle mud flap arrangement **10** generally comprises a drive unit **12**, an axle **14**, mud flaps **16**, and a control unit **18**. The drive unit **12** is designed to be mounted adjacent wheels **62** of a vehicle **60**. The axle **14** is operatively connected to the drive unit **12** and designed to be rotated by the drive unit **12**. The mud flaps **16** are suspended from opposing ends of the axle **14**. The control unit **18** is operatively connected to the drive unit **12** and designed to control the operation of the drive unit **12**. The drive unit **12** is actuatable to rotate the drive axle **14** and roll up and retract the mud flaps **16** from an extended position to a retracted position to minimize contact with objects **64** on the ground and damage resulting from such contact when a vehicle **60** is driven in reverse.

[0021] In accordance with at least one possible embodiment, the control unit **18** is connectable to a reverse indicator or reverse switch **66** of a vehicle **60** to automatically actuate the drive unit **12** to retract the mud flaps **16** when a vehicle **60** is driven in reverse.

[0022] In accordance with at least one possible embodiment, the control unit **18** is designed to actuate the drive unit **12** and rotate the axle **14** in a first direction, such as a clockwise or counterclockwise direction, until the mud flaps **16** are retracted to the retracted position in which the lower edge **20** of each of the mud flaps **16** is positioned about halfway up the height of its adjacent tire **62**. The control unit **18** is also designed to actuate the drive unit **12** and rotate the axle **14** in a second direction opposite the first direction until the mud flaps **16** are extended to the extended position in which the lower edge **20** of each of the mud flaps **16** is positioned three to six inches from the ground. FIG. **4** shows an example of the extended position in which the lower edge **20** is three inches from the ground. The dotted lines show the trajectory of objects, such as rocks, that could be propelled by the tires **62**, both at a three-inch height and a six-inch height. FIGS. **5** and **6** show an example of the retracted position in which the mud flaps **16** have been retracted such that the lower edge **20** is about at the halfway or midway point of the tire **62**. As shown in FIG. **6**, this position allows the mud flaps **16** to clear an object **64** on the ground, such as a parking block or curb. It should be understood that the retracted position could be a position in which the lower edge **20** is higher or lower off of the ground.

[0023] In accordance with at least one possible embodiment as shown in FIG. **7**, each of the mud flaps **16** includes an upper mounting bracket **22** and a flap portion **24** suspended in a freely pivotable or freely swinging manner from the upper mounting bracket **22**. The axle **14** also includes a mounting arrangement designed to receive and hold the upper mounting bracket **22** of each of the mud flaps **16** such that upon rotation of the axle **14** the flap portion **24** is rolled up around the axle **14**.

[0024] In accordance with at least one possible embodiment, the mounting arrangement includes an open-ended slot **26** in the drive axle **14** to permit sliding in and out of the upper mounting bracket **22** to permit insertion and removal of the mud flaps **16**, as well as a securing bolt **28** designed to be inserted into a receptacle **30** in the axle **14** to cover the open-ended slot **26** to secure the upper mounting bracket **22** in the open-ended slot **26**.

[0025] In accordance with at least one possible embodiment, the retractable vehicle mud flap arrangement **10** further includes a status indicator **32** operatively connected to the control unit **18** and designed to be mounted on a surface in a cab of a vehicle **60** and to indicate whether the mud flaps **16** are in the extended position or the retracted position.

[0026] In accordance with at least one possible embodiment, the retractable vehicle mud flap arrangement **10** further includes a manual control button arrangement, which could be part of the status indicator **32**, operatively connected to the control unit **18** to permit manual actuation of the drive unit **12** to permit a user to selectively retract and extend the mud flaps **16**.

[0027] In accordance with at least one possible embodiment, the control unit **18** is designed to actuate the drive unit **12** to automatically extend the mud flaps **16** upon expiration of a predetermined amount of time. In other words, once the mud flaps **16** are retracted due to reverse driving or similar situation, it is necessary to extend them again when normal forward driving resumes. A timer can be set to a time that is sufficient to allow a driver to complete a reverse driving action with the mud flaps **16** retracted and clear of any danger. Once the timer is up and the reverse driving action has presumably been completed, the mud flaps **16** automatically extend so the vehicle **60** is ready to be driven again safely.

[0028] In accordance with at least one possible embodiment, the drive unit **12** includes a battery or is connectable to a battery **68** of a vehicle **60** to supply power to the drive unit **12**.

[0029] In accordance with at least one possible embodiment, each of the mud flaps is a standard hinged mud flap, as shown in FIG. **1**, or a weighted steel plate suspended from chain link material, as shown in FIG. **2**.

[0030] In order to protect mud flaps **16** using the retractable vehicle mud flap arrangement **10**, first the drive unit **12**, the axle **14**, and the control unit **18** must be mounted on a vehicle **60**. The mud flaps **16** are suspended from opposing ends of the axle **14**, which can be done before or after mounting the axle **14**. A user can then drive his vehicle **60** normally. However, when driving the vehicle **60** in reverse, the drive unit **12** is actuated to rotate the drive axle **14**, which in turn rolls up and retracts the mud flaps **16** from the extended position to the retracted position. In this manner the user can minimize contact of the mud flaps **16** with objects **64** on the ground, and thereby minimize or prevent damage resulting from such contact when the vehicle **60** is driven in reverse.

[0031] With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

[0032] Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. A retractable vehicle mud flap arrangement comprising: a drive unit being configured to be mounted adjacent wheels of a vehicle; an axle being operatively connected to said drive unit and configured to be rotated by said drive unit; mud flaps being suspended from opposing ends of said axle; a control unit being operatively connected to said drive unit and configured to control the operation of said drive unit; and said drive unit being actuatable to rotate said drive axle and roll up and retract said mud flaps from an extended position to a retracted position to minimize contact with objects on the ground and damage resulting from such contact upon a vehicle being driven in reverse.
2. The retractable vehicle mud flap arrangement of claim 1, wherein said control unit is connectable to a reverse indicator or reverse switch of a vehicle to automatically actuate said drive unit to retract said mud flaps upon a vehicle being driven in reverse.
3. The retractable vehicle mud flap arrangement of claim 2, wherein: said control unit is configured to actuate said drive unit and rotate said axle in a first direction until said mud flaps are retracted to said retracted position in which the lower edge of each of said mud flaps is positioned about halfway up the height of its adjacent tire; and said control unit is configured to actuate said drive unit and rotate said axle in a second direction opposite said first direction until said mud flaps are extended to said extended position in which the lower edge of each of said mud flaps is positioned three to six inches from the ground.
4. The retractable vehicle mud flap arrangement of claim 3, wherein: each of said mud flaps comprises an upper mounting bracket and a flap portion suspended in a freely pivotable manner from said upper mounting bracket; and said axle comprises a mounting arrangement configured to receive and hold said upper mounting bracket of each of said mud flaps such that upon rotation of said axle said flap portion is rolled up around said axle.
5. The retractable vehicle mud flap arrangement of claim 4, wherein said mounting arrangement comprises: an open-ended slot in said drive axle to permit sliding in and out of said upper mounting bracket to permit insertion and removal of said mud flaps; and a securing bolt configured to be inserted into a receptacle in said axle to cover said open-ended slot to secure said upper mounting bracket in said open-ended lot.
6. The retractable vehicle mud flap arrangement of claim 5, further comprising a status indicator operatively connected to said control unit and configured to be mounted on a surface in a cab of a vehicle and to indicate whether said mud flaps are in said extended position or said retracted position.
7. The retractable vehicle mud flap arrangement of claim 6, further comprising a manual control button arrangement operatively connected to said control unit to permit manual actuation of said drive unit to permit a user to selectively retract and extend said mud flaps.
8. The retractable vehicle mud flap arrangement of claim 7, wherein said control unit is configured to actuate said drive unit to automatically extend said mud flaps upon expiration of a predetermined amount of time.
9. The retractable vehicle mud flap arrangement of claim 8, wherein said drive unit comprises a battery or is connectable to a battery of a vehicle to supply power to said drive unit.
10. The retractable vehicle mud flap arrangement of claim 9, wherein each of said mud flaps comprises a standard hinged mud flap or a weighted steel plate suspended from chain link material.
11. The retractable vehicle mud flap arrangement of claim 1, wherein: said control unit is configured to actuate said drive unit and rotate said axle in a first direction until said mud flaps are retracted to said retracted position in which the lower edge of each of said mud flaps is positioned about halfway up the height of its adjacent tire; and said control unit is configured to actuate said drive unit and rotate said axle in a second direction opposite said first direction until said mud flaps

are extended to said extended position in which the lower edge of each of said mud flaps is positioned three to six inches from the ground.

12. The retractable vehicle mud flap arrangement of claim 1, wherein: each of said mud flaps comprises an upper mounting bracket and a flap portion suspended in a freely pivotable manner from said upper mounting bracket; and said axle comprises a mounting arrangement configured to receive and hold said upper mounting bracket of each of said mud flaps such that upon rotation of said axle said flap portion is rolled up around said axle.

13. The retractable vehicle mud flap arrangement of claim 12, wherein said mounting arrangement comprises: an open-ended slot in said drive axle to permit sliding in and out of said upper mounting bracket to permit insertion and removal of said mud flaps; and a securing bolt configured to be inserted into a receptacle in said axle to cover said open-ended slot to secure said upper mounting bracket in said open-ended slot.

14. The retractable vehicle mud flap arrangement of claim 1, further comprising a status indicator operatively connected to said control unit and configured to be mounted on a surface in a cab of a vehicle and to indicate whether said mud flaps are in said extended position or said retracted position.

15. The retractable vehicle mud flap arrangement of claim 1, further comprising a manual control button arrangement operatively connected to said control unit to permit manual actuation of said drive unit to permit a user to selectively retract and extend said mud flaps.

16. The retractable vehicle mud flap arrangement of claim 1, wherein said control unit is configured to actuate said drive unit to automatically extend said mud flaps upon expiration of a predetermined amount of time.

17. The retractable vehicle mud flap arrangement of claim 1, wherein said drive unit comprises a battery or is connectable to a battery of a vehicle to supply power to said drive unit.

18. The retractable vehicle mud flap arrangement of claim 1, wherein each of said mud flaps comprises a standard hinged mud flap or a weighted steel plate suspended from chain link material.

19. A method of protecting mud flaps using the retractable vehicle mud flap arrangement of claim 1 comprising the steps of: mounting said drive unit, said axle, and said control unit on a vehicle; suspending said mud flaps from opposing ends of said axle; and driving said vehicle in reverse and actuating said drive unit and rotating said drive axle and rolling up and retracting said mud flaps from an extended position to a retracted position to minimize contact with objects on the ground and damage resulting from such contact upon said vehicle being driven in reverse.
